

Project Phase 2 Report

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Title: Consultancy Appointment and Management System (CAMS)

1. Relational Schema

Strong Entities: Client, Professional, Service, Location, Appointment

Weak Entities: Payment, Review

Client{

Client_id (Primary Key)

Name (NOT NULL)

Email (NOT NULL)

Phone

Date_of_birth

}

Professional{

Professional_id (Primary Key)

Name (NOT NULL)

Years_of_experience

License_number (NOT NULL)

Specialization (NOT NULL)

}

Service{

Service_id (Primary Key)

Description

Service_name (NOT NULL)

}

Location{

Location_id (Primary Key)

Address (NOT NULL)

Location_type

}

Appointment{

Appointment_id (Primary Key)

date (NOT NULL)

Client_id (Foreign Key referencing client, NOT NULL) - Maps BOOKS relationship

Professional_id (Foreign Key referencing professional, NOT NULL) - Maps CONDUCTS

relationship

Service_id (Foreign Key referencing service, NOT NULL) - Maps FOR relationship

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        Location_id (Foreign Key referencing location, NOT NULL) - Maps TAKES PLACE AT
relationship
    }

```

```

Payment{
    Payment_id (Partial Key)
    Appointment_id (Foreign Key referencing Appointment)
    Primary Key: (payment_id, appointment_id)

    Amount (NOT NULL)
    Payment_method
    Payment_status (NOT NULL)
    Payment_date (NOT NULL)
}

```

```

Review{
    Review_id (Partial Key)
    Appointment_id (Foreign Key referencing Appointment)
    Primary Key: (review_id, appointment_id)

    Rating (NOT NULL)
    Comment
}

```

Relational Schema Mapping Explanation

- Mapping of Strong Entities**
 Client, Professional, Service, Location, and Appointment were mapped as independent tables in the relational schema because they represent strong entities in the ER diagram. Each of these entities has its own primary key attribute (client_id, professional_id, service_id, location_id, and appointment_id), which uniquely identifies each record in the corresponding table.
- All attributes defined in the ER model were converted into columns with appropriate SQL data types. Mandatory attributes were implemented using **NOT NULL** constraints to ensure that required information must be provided when inserting records.
- Mapping of Relationships through the Appointment Table**
 The Appointment entity functions as the central component of the system and connects the main entities together. In the ER diagram, the relationships BOOKS (Client–Appointment), CONDUCTS (Professional–Appointment), FOR (Appointment–Service), and TAKES PLACE AT (Appointment–Location) are modeled as **one-to-many (1:N)** relationships.
- To implement these relationships in the relational model, the primary keys of Client, Professional, Service, and Location were added to the Appointment table as **foreign keys**. This design ensures that each appointment is associated with exactly one client, one professional, one service, and one location, while each of these entities can be linked to multiple appointments.

- Since the business rules specify that every appointment must be related to these entities, the foreign key attributes were defined with **NOT NULL** constraints to enforce total participation.
- **Mapping of Weak Entities**
Payment and Review were modeled as weak entities because they depend on the existence of an Appointment and cannot exist independently.
- To represent this dependency, the **appointment_id** attribute was included in both the Payment and Review tables as a **foreign key referencing Appointment**. In addition, appointment_id and payment_id together form the composite primary key of the Payment table, and appointment_id and review_id together form the composite primary key of the Review table. This design ensures that each appointment can have at most one associated payment record and one review record.
- The attributes payment_id and review_id, which served as partial keys in the ER diagram, were retained as additional columns in the relational tables to store identification information specific to each payment and review record

AI Tool Used: Claude

Prompt Used: “Here is our Phase 2 report for a database systems course. We have written the relational schema and mapping explanation based on our Phase 1 ER diagram. Please review it for correctness for example, check if the primary keys, foreign keys, NOT NULL constraints, and weak entity mappings are accurate and consistent with standard ER-to-relational mapping rules.”

AI Response: “The composite primary key for the Payment and Review tables was defined twice in the schema — once inside the entity block and once outside. Additionally, the mapping explanation incorrectly stated that appointment_id alone served as the primary key of these tables. The correct primary keys are composite: (payment_id, appointment_id) for Payment and (review_id, appointment_id) for Review.” (The corrected wording was provided and applied to the report.)